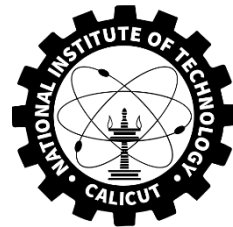


Quality, Reliability & Maintenance

ME3327E



तमसो मा ज्योतिर्गमय

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Quality Cost

Prevention Costs

- Quality planning and engineering
- New products review
- Product/process design
- Process control
- Burn-in
- Training
- Quality data acquisition and analysis

Appraisal Costs

- Inspection and test of incoming material
- Product inspection and test
- Materials and services consumed
- Maintaining accuracy of test equipment

Internal Failure Costs

- Scrap
- Rework
- Retest
- Failure analysis
- Downtime
- Yield losses
- Downgrading (off-specing)

External Failure Costs

- Complaint adjustment
- Returned product/material
- Warranty charges
- Liability costs
- Indirect costs

Measuring Quality Costs

- The magnitude of quality costs is important to management because such indices as return on investment are calculated from it.
- For comparing quality costs over time, magnitude may not be the measure to use because conditions often change from one quarter to the next.
- The number of units produced may change, which affects the direct costs of labor and materials, so the total cost in Rupees/dollars may not be comparable.
- To alleviate this situation, a measurement base that accounts for labor hours, manufacturing costs, sales Rupees/dollars, or units produced could be used to produce an index.
 - Labor-based index
 - Cost-based index
 - Sales-based index
 - Unit-based index

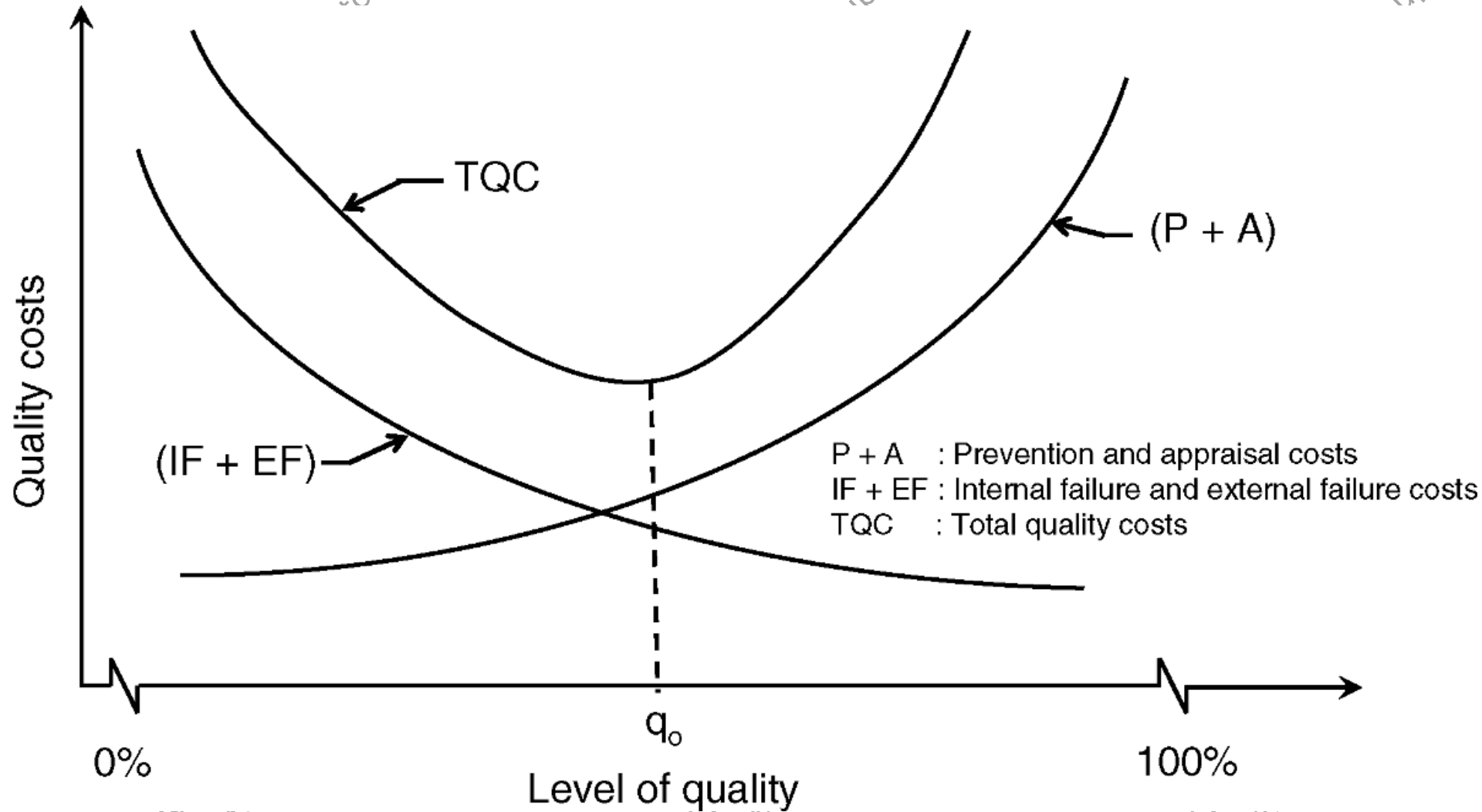
Exercise

A medical industry manufacturer of blood filtration products, used in dialysis, has to meet rigid standards in the field. Product design, precision manufacturing equipment, and raw material/component quality from suppliers are all important. Product failure at the customer level is quite expensive and could cost ₹ 25,00,000 for each occurrence. Yearly expenditures for the past year are shown in Table 3. Analyze the various quality cost components. What proportion are they of total quality costs? Find the total quality cost as a proportion of manufacturing costs and sales and comment. Should senior management stress less on design issues?

Table 3 Quality Cost Expenditures

Cost Categories	Amount (₹)
Review and update of design	2,00,00,000
Planning for quality through quality improvement teams	50,00,000
Operator training	20,00,000
Maintenance of ISO certification	1,00,00,000
Vendor selection	20,00,000
Incoming raw material inspection	30,00,000
Calibration of test equipment	10,00,000
In-house product testing	40,00,000
Scrap costs of non conforming product	50,00,000
Failure within warranty period	1,00,00,000
Manufacturing costs	700,00,00,000
Sales	1000,00,00,000

Impact of Quality Improvement on Quality Costs

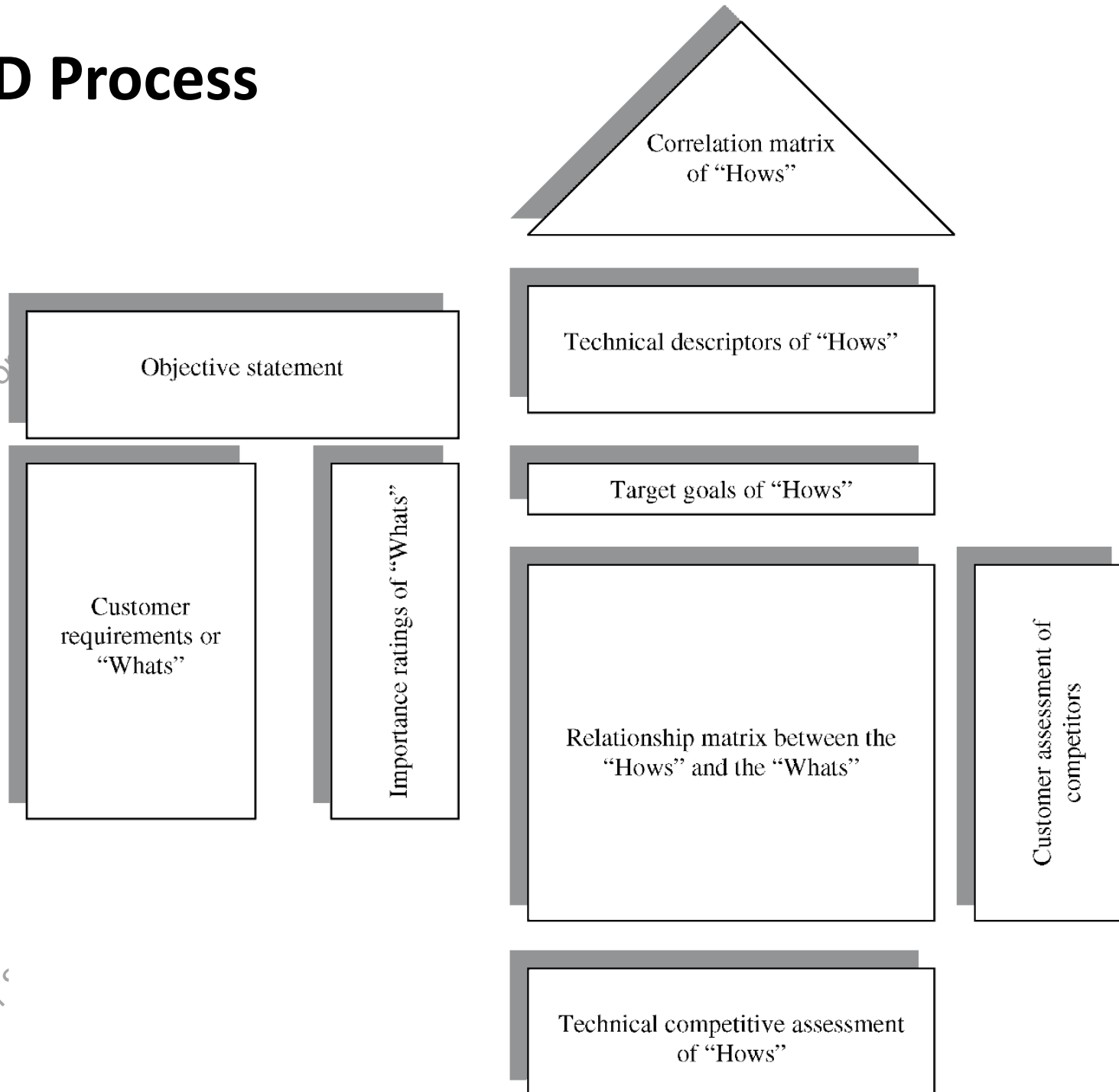


Quality costs versus level of quality

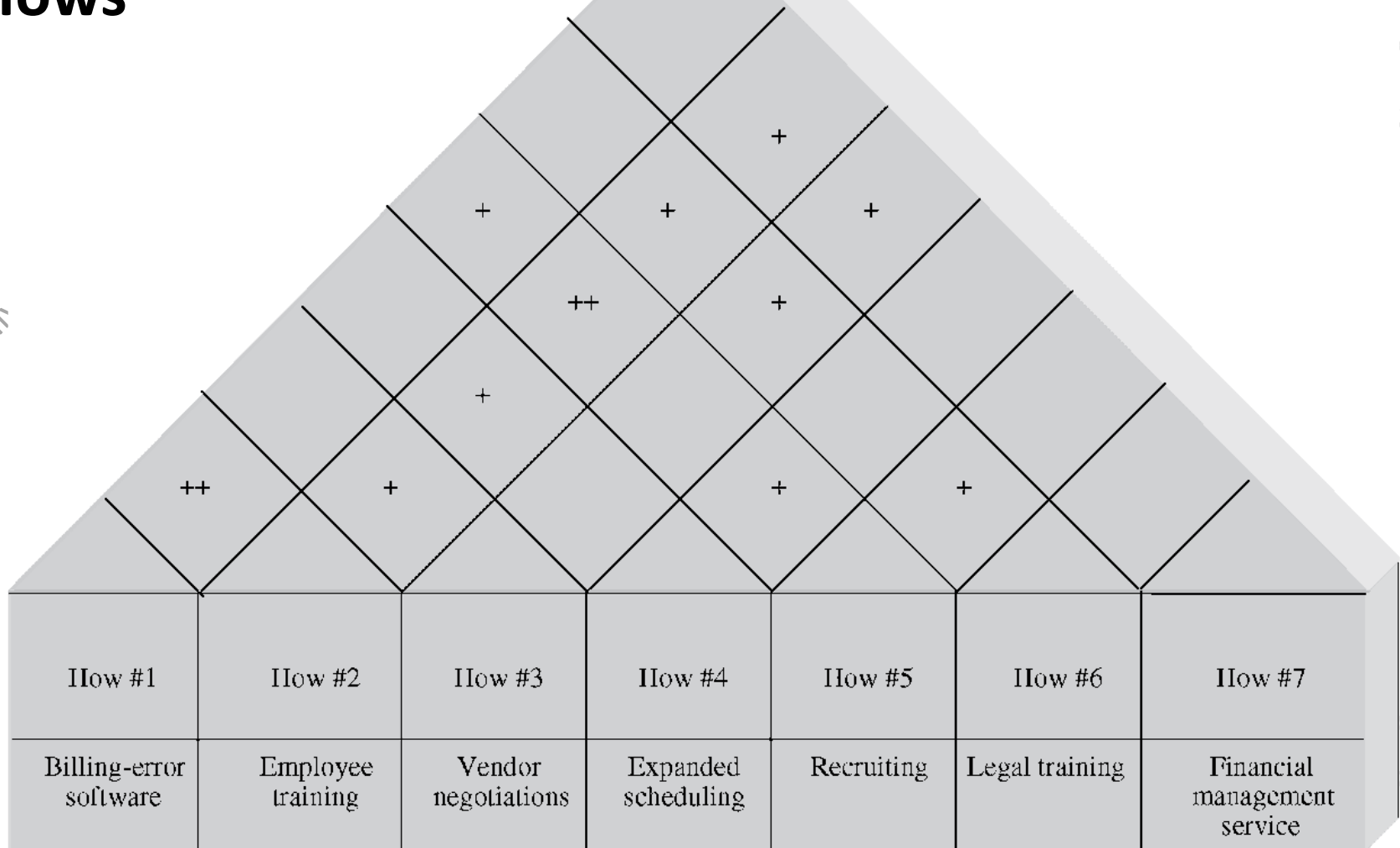
Quality Function Deployment

- QFD is a planning tool that focuses on designing quality into a product or service by incorporating customer needs.
- It is a systems approach involving cross-functional teams (whose members are not necessarily from product design) that looks at the complete cycle of product development.
- This quality cycle starts with creating a design that meets customer needs and continues on through conducting detailed product analyses of parts and components to achieve the desired product, identifying the processes necessary to make the product, developing product requirements, prototype testing, final product or service testing, and finishing with after-sales troubleshooting.
- QFD reduces the product development cycle time in each functional area, from product inception and definition to production and sales.

QFD Process



Correlation matrix of “Hows”



Importance Rating of Credit-Card Customer Requirements and Customer Assessment of Competitors

Customer Requirement (“Whats”)	Competitive Assessment of Companies			
	A	B	C	D
Low interest rate	2	2	4	2
Error-free transactions	5	5	3	3
No annual fee	1	5	2	3
Extended warranty at no additional cost	3	2	1	4
Customer service 24 hours a day	4	2	4	3
Customers’ advocate in billing disputes	4	2	3	3

Target Goals of Technical Descriptors

“Hows”	1	2	3	4	5	6	7
Target goals	⊙	↑	↑	⊙	↑	↑	↑

Legend

Number	Technical descriptors or “hows”
1	Software to detect billing errors
2	Employee training on data input and customer services
3	Negotiations with manufacturers and retailers (vendors)
4	Expanded scheduling (including flextime) of employees
5	Effective recruiting
6	Legal training
7	Financial management services
Symbol	Target goal
↑	Maximize or increase attained value
↓	Minimize or decrease attained value
⊙	Achieve a target value

Quality Loss Function

- The Quality Loss Function can be defined as a mathematical model that measures the financial loss incurred due to deviations from customer requirements or specifications.
- The function takes into account factors such as customer dissatisfaction, rework costs, warranty claims, and lost sales opportunities.
- The relevance of the Quality Loss Function in Quality Management is evident in its ability to identify areas where improvements can be made to reduce defects and enhance customer satisfaction.
- By understanding the financial implications of quality issues, organizations can prioritize their efforts and allocate resources effectively to address the most critical areas of concern.

Significance of quality loss function

- It helps organizations to identify the root causes of quality defects and take appropriate corrective actions.
- It enables organizations to set quality targets and measure their progress towards achieving them.
- It provides a common language for communication within the organization and with external stakeholders.
- It serves as a valuable tool for Quality Management businesses to drive continuous improvement, enhance customer satisfaction, and reduce costs associated with quality defects.

Benefits from quality loss function

- **Organizations:** By implementing the Quality Loss Function, organizations can identify areas of improvement, reduce defects, and ultimately enhance customer satisfaction.
- **Customers:** The Quality Loss Function helps organizations prioritize customer requirements and ensure that products or services meet their expectations.
- **Employees:** By reducing quality defects, organizations create a better working environment for employees.
- **Suppliers:** The Quality Loss Function encourages collaboration and communication between organizations and their suppliers.
- **Shareholders:** Shareholders benefit from the implementation of the Quality Loss Function through increased profitability and improved return on investment.

Best practices

- **Data Collection and Analysis:** Organizations should establish robust data collection processes to gather relevant information on quality defects, customer complaints, and other quality-related metrics.
- **Cross-functional Collaboration:** Quality Management is a cross-functional endeavor that requires collaboration between different departments and stakeholders.
- **Continuous Improvement:** The Quality Loss Function should be viewed as a continuous improvement tool.
- **Supplier Engagement:** Organizations should involve their suppliers in quality improvement efforts.
- **Leadership Support:** Top management support is critical for the successful implementation of the Quality Loss Function.

Product Control Model

- Product control is another method of statistical quality control in which the quality of a product is controlled while the product is ready to dispatch or sell to the customers. Product control used the technique of acceptance sampling to detect defects and control the quality of a product.

Statistical Concept and Techniques

Terms

- Population: Set of all items that possess a certain characteristics of interest
- Sample: Subset of Population
- Parameters: Characteristics of Population
- Statistics (Estimator): Characteristics of Sample
- Probability: Chance of occurrence of the events; $P(A) = \frac{n_A}{N}$
- Simple events: Cannot be broken down into other events
- Compound events: Made up of 2 or more events
- Complimentary events: Complimentary of events; $P(A^c) = 1 - P(A)$

Statistical Concept and Techniques

Terms

- Additive Law: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Multiplicative Law: $P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$
- Statistics: Science that deals with collection, analysis
 - Descriptive statistics
 - Inferential statistics
- Data collection: Variables
 - Continuous
 - Discrete
- Accuracy: Degree of uniformity
- Precision: Degree of Variability

Statistical Concept and Techniques

Terms

- Measurement Scales:

- Nominal scale
- Ordinal scale
- Interval scale
- Ratio scale

- Measure of central tendency:

- Mean: $\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$ (Sample); $\mu = \frac{\sum_{i=1}^N X_i}{N}$ (Population)
- Median: Value in the middle when observation is ranked
- Mode: Most frequently value occurred
- Trimmed mean

Statistical Concept and Techniques

Terms

- Measures of Dispersion (Variability):

- Range

- Variance: $\sigma^2 = \frac{\sum_{i=1}^N (X_i - \mu)^2}{N}$ (Population); $s^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$ (Sample)

- Standard variation

- Interquartile range (IQR): $Q_3 - Q_1$

- Skewness Coefficient: $\gamma_1 = \frac{n}{(n-1)(n-2)} \frac{\sum_{i=1}^n (X_i - \bar{X})^3}{s^3}$

- Kurtosis Coefficient:

$$\gamma_2 = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \frac{\sum_{i=1}^n (X_i - \bar{X})^4}{s^4} - \frac{3(n-1)^3}{(n-2)(n-3)}$$

Statistical Concept and Techniques

Terms

- Measures of Association:
 - Correlation coefficient:

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$